# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 07

**Full Syllabus (Chapters 1–13)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours 15 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** Grade 8 science combines wonder and evolution into: 1

**(a)** copying **(b)** investigation **(c)** recitation **(d)** memorisation

**2.** Even the simple puffing of a puri is: 1

**(a)** impossible to observe **(b)** untestable

**(c)** not completely understood by scientists **(d)** fully solved

**3.** The part of a cell that controls its activities and holds genetic material is the: 1

**(a)** nucleus **(b)** vacuole **(c)** cell membrane **(d)** cell wall

**4.** The first antibiotic, penicillin, was discovered by: 1

**(a)** Edward Jenner **(b)** Louis Pasteur

**(c)** Kamal Ranadive **(d)** Alexander Fleming

**5.** Ascariasis is caused by: 1

**(a)** roundworms **(b)** bacteria **(c)** viruses **(d)** protozoa

**6.** A coil of wire carrying current behaves like a: 1

**(a)** magnet **(b)** resistor only **(c)** switch **(d)** insulator

**7.** Which wire has higher resistance for the same size and length? 1

**(a)** silver **(b)** copper **(c)** aluminium **(d)** nichrome

**8.** Which of the following is a contact force? 1

**(a)** gravitational force **(b)** friction

**(c)** magnetic force **(d)** electrostatic force

**9.** A force of 900 N acts perpendicularly on an area of 3 $m^2$. The pressure is: 1

**(a)** 2700 $N/m^2$ **(b)** 300 $N/m^2$ **(c)** 150 $N/m^2$ **(d)** 600 $N/m^2$

**10.** A cyclone weakens after reaching land because: 1

**(a)** wind turns stronger **(b)** moist air supply is cut off

**(c)** land is cold **(d)** ocean returns

**11.** Increasing temperature of a gas generally makes its particles: 1

**(a)** move faster **(b)** stop **(c)** move slower **(d)** stick

**12.** Stainless steel is stronger than pure iron because it is a: 1

**(a)** mixture (alloy) **(b)** gas **(c)** pure element **(d)** compound

**13.** Which statement is correct about compounds? 1

**(a)** They can be separated by filtration
**(b)** They are always mixtures
**(c)** They have variable composition
**(d)** They have properties different from their elements

**14.** A gas–liquid solution is: 1

**(a)** soda water **(b)** brass **(c)** sugar syrup **(d)** air

**15.** Convex mirrors are preferred for traffic observation because they: 1

**(a)** enlarge the images **(b)** invert the images
**(c)** give a larger view area **(d)** absorb light

**16.** The Gregorian calendar is a: 1

**(a)** lunar calendar **(b)** sidereal lunar calendar
**(c)** solar calendar **(d)** luni-solar calendar

**17.** A sequence showing who eats whom is a: 1

**(a)** food chain **(b)** habitat **(c)** ecosystem **(d)** food web

**18.** Kunapa Jala in Vrikshayurveda is a: 1

**(a)** seed
**(b)** liquid fertiliser from animal and plant waste
**(c)** weedicide
**(d)** pesticide

**19.** The habitable zone is also called the: 1

**(a)** Hill zone **(b)** Roche zone **(c)** Goldilocks zone **(d)** Van Allen zone

**20.** Plants like sugarcane and rose can grow from: 1

**(a)** stem cuttings **(b)** bark only **(c)** leaf shape **(d)** pollen only

## SECTION — B (Very Short Answer Type) 8 × 2 = 16

*Answer in about 10–15 words each.*

**21.** State why cells differ in shape. 2

**22.** Mention two lifestyle habits that help prevent non-communicable diseases. 2

**23.** What is a voltaic cell? 2

**24.** What is the eye of a cyclone? 2

**25.** Why is air considered a mixture? 2

**26.** A cube has mass 54 g and volume 20 $cm^3$. Find the density. 2

**27.** Mention two human activities that threaten ecosystems. 2

**28.** Give two examples of plants that reproduce asexually. 2

## SECTION — C (Short Answer Type) 8 × 3 = 24

*Answer in about 30–40 words each.*

**29.** A student says 'hot oil makes puris puff better' without trials. Explain what is missing. 3

**30.** Why is it easier to slip on a wet floor? 3

**31.** Why can a gas be filled in a container of any size? 3

**32.** Explain the term ‘field of view’ and which mirror gives the largest. 3

**33.** Why is space debris a problem and what is being done? 3

**34.** What would happen to Earth if its atmosphere escaped? 3

**35.** Explain the working principle of a voltaic cell. 3

OR

Why should plugs and wires of proper rating be used? 3

**36.** Explain how decomposers recycle nutrients. 3

OR

Why are the Sundarbans considered a threatened ecosystem? 3

## SECTION — D (Long Answer Type) — 4 × 5 = 20

*Answer in about 80–100 words each.*

**37.** What are non-communicable diseases? Describe their causes, examples and ways to prevent them. 5

OR

Describe the types of diseases, how communicable diseases spread, and three precautions. 5

**38.** Describe the destructive effects of cyclones and the precautions that can reduce loss. 5

OR

Compare a thunderstorm and a cyclone on formation, location, effects and duration. 5

**39.** Define density. Describe an experiment to find the density of a stone or a small block. 5

OR

Explain the effect of temperature on solubility of solids and gases with two examples each. 5

**40.** Explain biotic and abiotic components, and describe how they interact in a pond ecosystem. 5

OR

Describe food chain and food web. Draw a food web with at least six organisms. 5

## END OF QUESTION PAPER — Total: 80 Marks

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